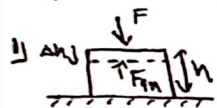


Elasticity

Stress & Strain in an elastic material:

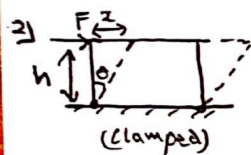
Internal force opposing the external deforming force per unit area

→ In deformed state, generally the internal force is equal to external force at equilibrium state.



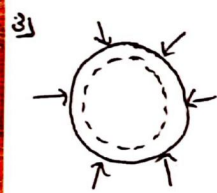
$$\text{Normal strain} = \frac{\Delta h}{h}$$

$$\text{Normal stress} = F/A$$



$$\text{Shear strain } (\theta) = \frac{x}{h}$$

$$\text{Shear stress} = F/A$$

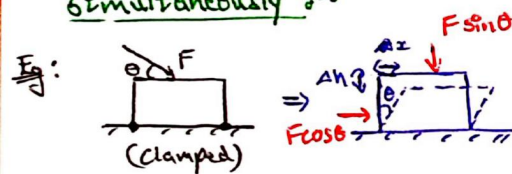


Volume of a liquid or gas is reduced by external pressure.

$$\text{Volume strain} = \frac{\Delta V}{V}$$

$$\text{Volume stress} = P_{ext.}$$

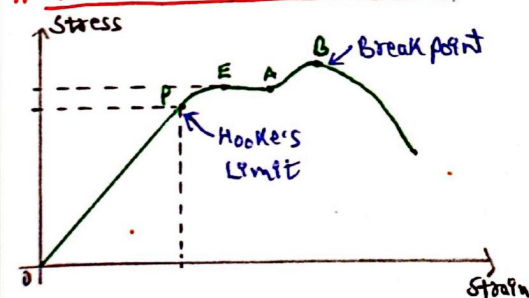
* Multiple types of strain acting simultaneously :-



$$\text{So, Normal strain} = \frac{F \sin \theta}{A}$$

$$\text{Shear strain} = \frac{F \cos \theta}{A}$$

Relation b/w stress & strain:



→ Stress is directly proportional to strain in the region of. This is Hooke's law.

Hooke's law: The ratio of stress & strain is constant for a material & equal to material's modulus of elasticity.

$$E = \frac{\text{stress}}{\text{strain}} = \frac{\text{stress}}{\text{strain}}$$

$$E_{solid} > E_{liq.} > E_{gas}$$

→ Beyond P, material starts losing its elastic property.

→ From E to A, strain increases even w/o applying more stress

→ After B, material loses its elasticity completely & becomes a plastic body.

Types of Modulus of Elasticity:

1) Young's Modulus

$$Y = \frac{\text{Longitudinal stress}}{\text{Longitudinal strain}} = \frac{F/A}{\Delta L/L}$$

2) Shear Modulus

$$\eta = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{F/A}{\theta}$$

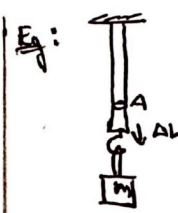
3) Bulk Modulus

$$\beta = \frac{\text{Volume stress}}{\text{Volume strain}} = \frac{\Delta P_{ext.}}{\Delta V/V}$$

[$\Delta P_{ext.}$, bcz in normal conditions $P_{ext.} = 1 \text{ atm}$]

Compressibility: Defined as the inverse of the Bulk modulus of substance.

$$C = 1/\beta$$

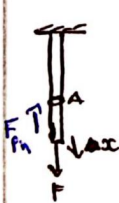


$$Y = \frac{Mg/A}{\Delta L/L}$$

$$\Rightarrow \Delta L = \frac{MgL}{YA}$$

Metal wires/Rods as Equivalent Springs:

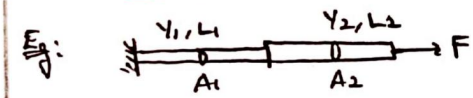
→ Metal rods/wires behaves as a spring & their various combinations can be solved like combination of springs.



$$\therefore Y = \frac{F/A}{\Delta L/L}$$

$$\Rightarrow F = \left(\frac{YA}{L}\right) \Delta L$$

$$\Rightarrow \text{Spring Const. } (K) = \frac{YA}{L}$$



$$\text{Soln } K_1 = \frac{Y_1 A_1}{L_1} \quad K_2 = \frac{Y_2 A_2}{L_2}$$

$$K_{eq} = \frac{K_1 K_2}{K_1 + K_2}$$

$$\text{So, } F = K_{eq} \cdot \Delta L_{total} \quad \underline{A_{net}}$$

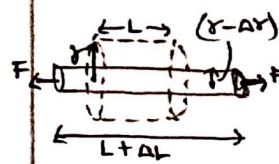
* PE stored in Rod :-

$$\text{Energy Density} = U_e = \frac{1}{2} \times \text{stress} \times \text{strain}$$

(Multiply by Vol. to get 'only' power)

Poisson's Ratio:

→ Defined for a body, whose deformation in length is so significant, that it also causes change in area at the same time.



$$\sigma = \frac{\text{Lateral strain}}{\text{Normal strain}}$$

$$\sigma = \frac{\Delta r/r}{\Delta L/L}$$